Assignment 6

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Please note that

- **Released date: April 07, Sunday.**
- **Due date: April 17, Wednesday, by 11:55 pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "123010XXX ASS6.pdf" (Your student ID + ASS No.). You may either typeset your answers directly using computers or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size to 10MB.)

Question 1. Let $L : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear transformation.

1. Show that $S : \mathbb{R}^n \to \mathbb{R}^m$ defined as $S(x) = L(x) + T(x)$ is a linear transformation.
2. Let A and B be the matrix representations of L and T respectively. Verify that $A + B$ is the matrix representation of S.

Solution

1. Consider $\alpha, \beta \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$. We have
   $L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$
   $T(\alpha x+\beta y) = \alpha T(x)+\beta T(y)$ because L and T are linear transformations.
   Therefore,
   $$\begin{aligned} S(\alpha x + \beta y) &= L(\alpha x + \beta y) + T(\alpha x + \beta y) \\ &= \alpha L(x) + \beta L(y) + \alpha T(x) + \beta T(y) \\ &= (\alpha L(x) + \alpha T(x)) + (\beta L(y) + \beta T(y)) \\ &= \alpha S(x) + \beta S(y) \end{aligned}$$
2. Consider $x \in \mathbb{R}^n$. We have $L(x) = Ax, T(x) = Bx$,
   $S(x) = L(x) + T(x) = Ax + Bx = (A + B)x$.
   So, $A + B$ is the matrix representation of S.

Question 2. Consider two linear transformations $S : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^p$. Define $L : \mathbb{R}^n \to \mathbb{R}^p$ as $L(x) = T(S(x)), x \in \mathbb{R}^n$.

1. Show that L is a linear transformation.
2. Let A and B be the matrix representation of S and T respectively.
   a) Justify why the matrix product BA is defined.
   b) Verify that BA is the matrix representation of L.

Solution

1. Consider $\alpha, \beta \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$. Let $u = S(x) \in \mathbb{R}^m$ and $v = S(y) \in \mathbb{R}^m$. We have:

   - $S(\alpha x + \beta y) = \alpha S(x) + \beta S(y) = \alpha u + \beta v$, because S is a linear transformation.
   - $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$, because T is a linear transformation.
   -
   $$L(\alpha x + \beta y) = T(S(\alpha x + \beta y)) = T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$
   $$= \alpha T(S(x)) + \beta T(S(y)) = \alpha L(x) + \beta L(y)$$

2. a) $S : \mathbb{R}^n \to \mathbb{R}^m$ with matrix representation A, so $A \in \mathbb{R}^{m \times n}$.
   $T : \mathbb{R}^m \to \mathbb{R}^p$ with matrix representation B, so $B \in \mathbb{R}^{p \times m}$.
   $B \in \mathbb{R}^{p \times m}$ and $A \in \mathbb{R}^{m \times n}$, so BA is defined, and $BA \in \mathbb{R}^{p \times n}$.

   b) Consider $x \in \mathbb{R}^n$. We have $u = S(x) = Ax \in \mathbb{R}^m$.
   $T(u) = Bu \in \mathbb{R}^p$.
   $L(x) = T(S(x)) = T(u) = Bu = B(Ax) = BAx$.

Question 3. Consider $A, B \in \mathbb{R}^{n \times n}$. Suppose that A and B are similar.

1. Show that A^2 and B^2 are similar.
2. Show that, for any integer $n \geq 3$, A^n and B^n are similar.
3. Show that, if $A^2 = I$, then $B^2 = I$.
4. Show that, if A is invertible, then B is invertible and A^{-1} and B^{-1} are similar.
5. Show that there is an invertible matrix S such that $Null(B) = \{x \in \mathbb{R}^n | \ \exists y \in Null(A), x = S^{-1}y\}$.
6. Show that $n(A) = n(B)$ and $rank(A) = rank(B)$.

Solution

$\exists S \in \mathbb{R}^{n\times n}$, S is invertible, and $B = S^{-1}AS$.

1. $B^2 = S^{-1}AS\ S^{-1}AS = S^{-1}AIAS = S^{-1}A^2S$.

2. By induction, if $B^n = S^{-1}A^nS$, then
   $B^{n+1} = S^{-1}A^nSS^{-1}AS = S^{-1}A^nIAS = S^{-1}A^{n+1}S$.

3. By [1.], if $A^2 = I$, then $B^2 = S^{-1}A^2S = S^{-1}IS = S^{-1}S = I$.

4. Suppose that A is invertible. Let $M = S^{-1}A^{-1}S^1$. Then
   $$\begin{aligned} MB &= S^{-1}A^{-1}SS^{-1}AS \\ &= S^{-1}A^{-1}IAS \\ &= S^{-1}A^{-1}AS \\ &= S^{-1}IS \\ &= S^{-1}S = I \end{aligned}$$
   So B is invertible, and $B^{-1} = M = S^{-1}A^{-1}S$. Thus A^{-1} and B^{-1} are similar.

5. Consider $x \in \mathbb{R}^n$. If $x \in Null(B)$, then $Bx = 0$, $S^{-1}ASx = 0$, multiply by S, $ASx = 0, Sx \in Null(A)$.
   Let $y = Sx$. Since S is invertible, $x = S^{-1}y$, with $y \in Null(A)$.
   If $x = S^{-1}y$ with $y \in Null(A)$, then,
   $Bx = S^{-1}ASx = S^{-1}ASS^{-1}y = S^{-1}AIy = S^{-1}Ay = 0$,
   so $x \in Null(B)$.

6. Let $\{v_1, v_2, ..., v_k\}$ be a basis of $Null(B)$. By part (5), each v_i can be written as $v_i = S^{-1}y_i$, where $y_i \in Null(A)$. Let $w_i = Sv_i$. Since S is invertible, the vectors $\{w_1, w_2, ..., w_k\}$ span the same space as $\{v_1, v_2, ..., v_k\}$.
   Moreover, $w_i = S(S^{-1}y_i) = y_i$, so $w_i \in Null(A)$.
   Thus, $dim\ Null(B) = k = dim\ Null(A)$, so $n(B) = n(A)$.
   By rank-nullity theorem,
   $n(B) + rank(B) = n$
   $n(A) + rank(A) = n$
   So $rank(B) = n - k = rank(A)$.

Question 4. Consider the linear transformation $L : P_3 \to P_2$ defined as
$L(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 + (2a_2 - a_1 - a_0)x + (2a_3 - 2a_1 - a_0)x^2$.

1. Find A, the matrix of L in the bases $B = \{1, x, x^2, x^3\}$ and $B' = \{1, x, x^2\}$.

2. Verify that, if $p \in Ker(L)$, then $x = [p]_B \in Null(A)$.
3. Find a basis of $Ker(L)$.
4. Verify that, if $q \in L(P_3)$, then $y = [q]_{B'} \in Col(A)$.
5. Find a basis of $L(P_3)$.

Solution

1. $B = \{p_1 = 1, p_2 = x, p_3 = x^2, p_4 = x^3\}$ is the standard basis of P_3.
   $B' = \{q_1 = 1, q_2 = x, q_3 = x^2\}$ is the standard basis of P_2.
   $L(p_1) = -x - x^2 = 0q_1 + (-1)q_2 + (-1)q_3$
   $L(p_2) = 1 - x - 2x^2 = 1q_1 - 1q_2 - 2q_3$
   $L(p_3) = 2x = 0q_1 + 2q_2 + 0q_3$
   $L(p_4) = 2x^2 = 0q_1 + 0q_2 + 2q_3$.
   $[L(p_1)]_{B'} = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}$, $[L(p_2)]_{B'} = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$, $[L(p_3)]_{B'} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$, $[L(p_4)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$.
   $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 0 \\ -1 & -2 & 0 & 2 \end{bmatrix}$.
2. Consider $p \in Ker(L)$. Then $L(p) = 0$. Let $x = [p]_B \in \mathbb{R}^4$. Then $[L(p)]_{B'} = Ax$, in which $L(p) = 0$. So $Ax = 0$. Thus $x \in Null(A)$.
3. We have $Ker(L) = \{p \in P_3 | x = [p]_B \in Null(A)\}$.
   If $x \in Null(A)$ and $x = [p]_B$, then $[L(p)]_{B'} = A[p]_B = Ax = 0$, so that $p \in Ker(L)$.

   To find a basis of $Null(A)$, transform A to REF:
   $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 0 \\ -1 & -2 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$
   The system $Ax = 0$ is equivalent to
   $\begin{cases} x_1 + x_2 - 2x_3 = 0 \\ x_2 = 0 \\ x_3 - x_4 = 0 \end{cases}$

By back substitution:

$$\begin{cases} x_1 = 2x_4 \\ x_2 = 0 \\ x_3 = x_4 \\ x_4 \in \mathbb{R} \end{cases}.$$

So, $Null(A) = \{x = \begin{bmatrix} x_1 = 2x_4 \\ x_2 = 0x_4 \\ x_3 = x_4 \\ x_4 \end{bmatrix} | x_4 \in \mathbb{R}\}$

$= \{x = x_4 \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix} | x_4 \in \mathbb{R}\}.$

Let vectors $u = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, then

$Null(A) = \{x = x_4 u | x_4 \in \mathbb{R}\} = Span(u)$.

$Ker(L) = Span(p)$, where $[p]_B = u$, i.e. $p = 2 + x^2 + x^3$.

Therefore, $Ker(L) = Span(2 + x^2 + x^3)$.

4. Consider $q \in L(P_3)$. Then, $\exists p \in P_3, q = L(p)$.
   Let $x = [p]_B \in \mathbb{R}^n$ and $y = [q]_{B'} \in \mathbb{R}^m$. Then $y = Ax$, since $[L(p)]_{B'} = A[p]_B$.
   The system $Ax = y$ is consistent, so $y \in Col(A)$.

5. We have $L(p_3) = \{q \in P_2 |\ [q]_{B'} \in Col(A)\}$.
   Verify that, if $y = [q]_{B'} \in Col(A)$, then $q \in L(p_3)$.
   Transform A to RREF

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 0 \\ \underline{-1} & \underline{-2} & \underline{0} & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ \underline{0} & \underline{0} & \underline{1} & -1 \end{bmatrix},$$

   Column with leading is correspond to the basis column of A.

$$\{ \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \} \text{ is a basis of } Col(A).$$

$\{-x-x^2, 1-x-2x^2, 2x\}$ is a basis of $L(P_3)$.

Question 5. Are the following mappings linear transformations?

a) $L: \mathbb{R}^2 \to \mathbb{R}^3, L(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}) = \begin{bmatrix} 0 \\ 0 \\ |x_1+x_2| \end{bmatrix}$.

b) $L: P_2 \to \mathbb{R}^2, L(a_0+a_1x+a_2x^2) = \begin{bmatrix} a_0a_1 \\ a_2 \end{bmatrix}$.

Solution

a) No, because $L(-\begin{bmatrix} 1 \\ 0 \end{bmatrix}) = L(\begin{bmatrix} -1 \\ 0 \end{bmatrix}) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and $-L(\begin{bmatrix} 1 \\ 0 \end{bmatrix}) = -\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. So, $L(-\begin{bmatrix} 1 \\ 0 \end{bmatrix}) \neq -L(\begin{bmatrix} 1 \\ 0 \end{bmatrix})$.

b) No, because $L(2+2x+2x^2) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, and $2L(1+x+x^2) = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, so $L(2+2x+2x^2) \neq L(1+x+x^2)$.

Question 6. Let $L: \mathbb{R}^3 \to \mathbb{R}^3$ be defined as $L(x) = Ax$, where $A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$.

Consider the basis of $\mathbb{R}^3$, $B = \{v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}\}$. Let $B_0 = \{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$.

1. Find the transition matrix from B to B_0.
2. Find by two different methods, the matrix of L with respect to the basis B.

Solution

1. $v_1 = 1e_1 + 1e_2 + 1e_3$
   $v_2 = 1e_2 + 2e_2 + 0e_3$.
   $v_3 = 0e_1 - 2e_z + 1e_3$

$S = [\text{transition matrix from } B \text{ to } B_0] = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix}$

2. $B = S^{-1}AS$ is the basis of L in B, where $S^{-1} = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix}$ is obtained by transforming $[A|S]$ to RREF.

   Consequently,

$$B = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

· (Method 2): $L(v_1) = Av_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0v_1 + 0v_2 + 0v_3$

$$L(v_2) = Av_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = v_2 = 0v_1 + 1v_2 + 0v_3$$

$$L(v_3) = Av_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} = v_3 = 0v_1 + 0v_2 + 1v_3$$

Thus, $[L(v_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $[L(v_2)]_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $[L(v_3)]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Therefore, $[L]_B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.